Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/29/2007
PI Name:	Conley, Catherine A Ph.D.		
Project Title:	A New Model for Understanding Spaceflight Induced Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology		
Space Biology Special Category:	None		
PI Email:	cconley@mail.arc.nasa.gov	**Fax:**	FY 650-604-3954
PI Organization Type:	NASA CENTER	**Phone:**	650-604-0234
Organization Name:	NASA - AMES Research Center		
PI Address 1:	SLR		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:	NOTE (June 2016): Dr. Conley is currently at NASA Headquarters		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	07/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NCE to 9/30/2007, per A. Chu 6/23/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jacobson, Lewis (University of Pittsburgh) Szewczyk, Nathaniel (University of Pittsburgh)		
Grant/Contract No.:	ARC Intramural PI ; NNA04CK22A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this project is to develop the nematode C. elegans as a small animal model for studying the molecular basis of spaceflight induced muscle atrophy. C. elegans is an established model both for studying the molecular basis of muscle atrophy on Earth and for studying spaceflight effects on physiology. The principal challenge addressed during this project period is the development, characterization and evaluation of a chemically defined liquid culture system that allows automation of C. elegans experimentation and analysis in flight. We have broken this work into three Aims for the current project period: 1) Adapt the C. elegans muscle protein degradation model to liquid culturing; 2) Validate liquid culturing of C. elegans in the spaceflight environment; 3) Determine biological responses of C. elegans in liquid culture to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	1) We have developed a system for the automated culture of the common laboratory animal C. elegans. The system can be used as an enviornmental biomonitor, in pharmaceutical discovery, and in compound toxicity testing. 2) Based upon our work with worm muscles, which lack satellite cells, we believe that altered muscle development may be a contributing factor to human muscle pathologies (in the form of altered development of satellite cells which fuse with developed muscle). If true, this finding opens new avenues of investigation toward muscle pathology treatments. 3) We have identified new genetic triggers of and signaling pathways which regulate protein degradation in worm muscle. These pathways may represent targets for new treatments of human muscle pathologies.
Task Progress:	We have successfully demonstrated that it is possible to culture the commonly employed laboratory organism C. elegans in flight using a chemically defined medium and autonomous culturing equipment. Specific findings/results of note include: 1) The first demonstration of normal animal growth, development, and behaviour following 10 generation in space. 2) The successful involvement of school children around the world in real time analysis of animal growth and behaviour onboard the ISS. 3) The demonstration that there appears to be a conserved molecular basis for muscular response to spaceflight in worms, rodents, and astronauts. 4) The identification of a novel, attachment based, mechanisms regulating muscle protein degradation.
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Abstracts for Journals and Proceedings	Stevenson T, Mercer K, Cox E, Szewczyk N, Conley C, Hardin J, Benian G. "unc-94 Encodes a Tropomodulin in C. elegans." 16th International C. elegans meeting, June 2007. 16th International C. elegans meeting proceedings, June 2007. , Jun-2007
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Abstracts for Journals and Proceedings	Szewczyk N. "Using transgenic C. elegans to identify and model the signaling networks controlling muscle proteolysis." Poster: National Institute of Arthritis and Musculoskeletal and Skin Diseases sponsored Workshop on Investigating Human Muscle Function in vivo, Vanderbilt University, October 2005. National Institute of Arthritis and Musculoskeletal and Skin Diseases sponsored Workshop on Investigating Human Muscle Function in vivo Workshop Proceedings, 2005. , Oct-2005
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